REPORT

DATA ANALYTICS INTERNSHIP

ShadowFox

Projects Covered:

- 🟢 **Beginner: Vrinda Store Sales Analysis — Excel**
- 🟡 **Intermediate: SuperStore Sales Analysis — Python & Power BI**
- 🔵 **Advanced: IBM HR Analytics — Python & Power BI**

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1. INTRODUCTION

The ShadowFox Data Analytics Internship provided an opportunity to develop practical skills in data analysis, visualization, dashboard development, and business-oriented interpretation through a progression of projects across three levels.

The internship was structured around increasing analytical complexity. The Beginner Level focused primarily on spreadsheet-based analysis and dashboard presentation. The Intermediate Level introduced structured data cleaning, exploratory analysis, and interactive Power BI reporting. The Advanced Level further developed these skills through the creation of an executive-style HR Analytics dashboard with KPIs, interactive filters, and business-focused insights.

The progression across the three projects provided practical experience in transforming raw datasets into meaningful information that can support business decisions.

The official ShadowFox task structure describes the Beginner level as focusing on basic data handling, spreadsheet analysis, chart usage, dashboard clarity, interpretation of trends, and summary reporting. The Intermediate level requires deeper analysis, data preparation, exploratory analysis, structured reporting, insights, and recommendations. The Advanced level focuses on executive-style Power BI/Tableau dashboards, KPI tracking, interactive reporting, visual storytelling, and decision-oriented recommendations.

2. INTERNSHIP OBJECTIVES

The major objectives of the internship were:

- To develop practical data analytics skills.
- To understand and work with real-world datasets.
- To perform data cleaning and preparation.
- To analyze business data and identify meaningful patterns.
- To create clear and effective data visualizations.
- To develop interactive dashboards.
- To select meaningful KPIs for business reporting.
- To interpret analytical results from a business perspective.
- To communicate findings through structured reports.
- To develop decision-oriented recommendations.
- To gain practical experience with professional data-analysis tools.

The internship also emphasized progression from simply presenting numbers toward understanding the business questions behind the data and communicating insights effectively.

3. TOOLS AND TECHNOLOGIES USED

3.1 Microsoft Excel

Microsoft Excel was primarily used for the Beginner Level project.

Key features included:

- Data organization
- Pivot Tables
- Pivot Charts
- Slicers
- Summary analysis
- Dashboard creation
- Data visualization

Excel provided the foundation for understanding how raw transactional data can be summarized and presented visually.

3.2 Python

Python was used in the Intermediate and Advanced projects for data preparation and analysis.

The primary library used was:

Pandas

Python was used for activities such as:

- Loading datasets
- Inspecting data
- Checking data types
- Identifying missing values
- Checking duplicate records
- Transforming data
- Creating derived fields
- Preparing cleaned datasets for Power BI

3.3 Google Colab

Google Colab was used as the working environment for Python-based data preparation.

It provided a convenient environment for:

- Running Python notebooks
- Loading CSV datasets
- Performing data cleaning
- Exploring datasets
- Creating transformed datasets
- Exporting cleaned data

3.4 Microsoft Power BI

Power BI was used for the Intermediate and Advanced dashboards.

Power BI enabled:

- KPI creation
- Interactive charts
- Slicers
- Dashboard design
- Business-focused visualization
- Comparative analysis
- Interactive filtering

The Advanced project particularly emphasized the use of Power BI to create an executive-style dashboard, matching the requirements of the Advanced internship level

3.5 Git and GitHub

Git and GitHub were used to organize and maintain the internship projects.

The repository contains the completed Beginner, Intermediate, and Advanced projects in separate folders, together with their datasets, notebooks, dashboards, and reports.

4. BEGINNER LEVEL – VRINDA STORE SALES ANALYSIS

4.1 Project Overview

The Beginner Level project focused on analyzing sales data for **Vrinda Store** using Microsoft Excel.

The primary objective was to understand sales performance, customer characteristics, order status, geographic contribution, and sales channels through an interactive Excel dashboard.

This project provided the foundation for developing basic data-analysis and visualization skills

4.2 Objectives

The main objectives of the project were:

- Analyze overall sales and order trends.
- Compare sales performance between women and men.
- Analyze order status.
- Identify the top-performing states.
- Examine the relationship between age group and gender.
- Analyze sales across different order channels.
- Present the results through an interactive dashboard.

4.3 Data Analysis

The project involved organizing the sales data and using Excel's analytical capabilities to summarize the information.

Pivot Tables were used to aggregate information, while Pivot Charts were used to present the results visually.

The dashboard included analysis of:

Sales and Order Trends

Sales and order activity were analyzed to understand overall performance and identify changes across the reporting period.

Gender Analysis

Sales were compared between women and men to understand the contribution of different customer groups.

Order Status

Order status was analyzed to understand the distribution of completed and other order outcomes.

Geographic Analysis

The top five states were identified based on sales contribution, helping highlight important geographic markets.

Age and Gender Analysis

Age groups were compared with gender to understand customer composition.

Sales Channels

Different order channels were analyzed to understand where sales were being generated.

4.4 Dashboard

The final Excel dashboard combined the major analytical components into a single interactive reporting environment.

Slicers were incorporated to allow users to filter the dashboard and explore different segments of the data.

This helped demonstrate the importance of interactivity in business reporting.

4.5 Learning Outcome

The Beginner project established fundamental skills in:

- Spreadsheet-based analysis
- Pivot Tables
- Pivot Charts
- Data summarization
- Dashboard organization
- Basic business interpretation
- Interactive filtering

It served as the starting point for the more advanced analytical techniques used in the subsequent projects.

5. INTERMEDIATE LEVEL – SUPERSTORE SALES ANALYSIS

5.1 Project Overview

The Intermediate project focused on the analysis of **SuperStore sales data**.

Unlike the Beginner project, this project introduced a dedicated data-cleaning stage using Python and Pandas before developing the Power BI dashboard.

The project therefore moved beyond basic spreadsheet reporting toward a more structured data-analysis workflow.

The ShadowFox Intermediate requirements emphasize realistic datasets, data cleaning and preparation, exploratory analysis, business/product/segment analysis, trend identification, visual reporting, insights, and recommendations.

5.2 Dataset

The SuperStore dataset used for the project contained approximately **5,901 records and 23 columns**.

The data contained information related to:

- Orders
- Customers
- Geography
- Products
- Categories
- Sub-categories
- Sales
- Quantity
- Profit
- Shipping
- Payment
- Returns

This provided a richer dataset compared with the Beginner project.

5.3 Data Cleaning

Python and Pandas were used to inspect and prepare the dataset.

The data-cleaning process included:

- Loading the raw CSV file.
- Inspecting the dataset structure.
- Checking column names.
- Checking data types.
- Examining missing values.
- Checking duplicate records.
- Inspecting categorical fields.
- Reviewing numerical fields.
- Preparing the cleaned dataset for Power BI.

The cleaning stage was important because reliable dashboards depend on correctly structured and validated data.

5.4 Exploratory Analysis

The project examined several dimensions of business performance.

Sales and Profit Trends

Sales and profit were analyzed over time to understand business performance and trends.

Category Analysis

Different product categories were compared to understand their contribution to sales and profitability.

Regional Analysis

Regional performance was examined to identify differences in sales and profit across geographic areas.

Customer Segment Analysis

Customer segments were compared to understand their relative contribution to business performance.

Product and Sub-category Analysis

Products and sub-categories were examined to identify strong and weaker areas of the product portfolio.

Returns Analysis

The dataset also contained return-related information, which was incorporated into the analysis to understand the impact of returned orders.

5.5 Power BI Dashboard

The cleaned dataset was imported into Power BI to create an interactive dashboard.

The dashboard incorporated:

- KPI cards
- Sales and profit trends
- Category analysis
- Regional analysis
- Product analysis
- Customer segment analysis
- Return analysis
- Interactive slicers

The use of Power BI made it possible to move from static analysis toward interactive business reporting.

5.6 Learning Outcome

The Intermediate project strengthened skills in:

- Python-based data preparation
- Pandas
- Exploratory data analysis
- Business performance analysis
- Power BI
- KPI reporting
- Interactive visualization
- Trend analysis
- Business insight generation

Compared with the Beginner project, the Intermediate project required greater attention to data quality, analytical structure, and business interpretation.

6. ADVANCED LEVEL – IBM HR ANALYTICS DASHBOARD

6.1 Project Overview

The Advanced project focused on the IBM HR Analytics dataset and the development of an executive-style HR Analytics Dashboard in Power BI.

The project was designed around employee workforce information and focused on areas such as employee attrition, job satisfaction, departments, job roles, overtime, age groups, and employee tenure.

The official ShadowFox Advanced requirements specifically allow the IBM HR Analytics dataset and require an interactive dashboard suitable for KPI tracking and business-ready storytelling.

6.2 Dataset

The IBM HR Analytics dataset used in the project contained:

- 1,470 employee records
- 35 columns

The dataset included employee-related information covering areas such as:

- Age
- Department
- Job Role
- Job Level
- Monthly Income
- Years at Company
- Total Working Years
- Job Satisfaction
- Overtime
- Attrition
- Gender
- Business Travel
- Education
- Marital Status
- Work-Life Balance

This dataset provided sufficient information to explore multiple dimensions of workforce behavior and employee attrition.

6.3 Data Preparation

Python and Pandas were used to prepare the dataset before importing it into Power BI.

The preparation process included:

1. Loading the raw dataset.
2. Inspecting its structure.
3. Checking data types.
4. Checking missing values.
5. Checking duplicate records.
6. Reviewing categorical values.
7. Examining numerical fields.
8. Identifying constant fields.
9. Creating age groups.
10. Creating salary groups.
11. Creating tenure groups.
12. Creating an Attrition Flag.
13. Creating an OverTime Flag.
14. Creating Company Tenure Percentage.
15. Exporting the cleaned dataset.

Importantly, the dataset did not require artificial filling of missing values or removal of duplicate records where none were present. The preparation focused on validation and useful transformation rather than claiming unnecessary cleaning operations.

6.4 KPI DEVELOPMENT

The Advanced dashboard contains six primary KPI cards:

1. Total Employees

Provides an overall view of the workforce represented in the dataset.

2. Attrition Count

Shows the number of employees classified as having left the organization.

3. Average Job Satisfaction

Provides a high-level measure of employee job satisfaction.

4. Average Age

Shows the average age of the employees represented in the dataset.

5. Average Monthly Income

Provides an overview of employee monthly income.

6. Average Years at Company

Shows the average organizational tenure of employees.

These KPIs provide an executive summary before users move into the detailed visual analysis.

6.5 DASHBOARD VISUALIZATIONS

The Advanced dashboard uses a focused single-page layout.

Attrition by Department

This visualization compares employee attrition across departments.

It helps HR stakeholders identify whether attrition is concentrated in particular organizational areas.

Job Role by Attrition

This visual examines attrition across different job roles.

It can help identify roles where employee turnover may require additional investigation.

Overtime by Attrition

This analysis compares attrition with overtime status.

It provides a useful view of whether employee turnover patterns differ between employees who work overtime and those who do not.

Age Group by Attrition

Employees are grouped into age categories and compared based on attrition.

This helps provide demographic context to employee turnover.

Average Years at Company by Attrition

This visual compares employee tenure across attrition categories.

It can help HR teams understand whether attrition patterns differ according to employee experience and length of service.

Employee Number by Department

This visualization provides a department-level view of employee records represented in the dataset.

Average Job Satisfaction by Department

This visual compares job satisfaction across departments and provides another perspective on workforce experience.

6.6 INTERACTIVE FILTERS

The dashboard includes interactive slicers for:

- Department
- Age Group
- Job Role
- Job Level
- OverTime

These filters allow users to move from a high-level workforce view to more specific employee segments.

For example, an HR stakeholder can filter a particular department and then examine job role, overtime, age group, and satisfaction patterns within that selection.

This supports the Advanced requirement for interactive reporting through filters and slicers.

6.7 BUSINESS INSIGHTS

The Advanced dashboard is designed to help HR stakeholders investigate several important questions:

- Which departments experience higher employee attrition?
- Which job roles show greater attrition?
- How does overtime relate to attrition?
- How does attrition vary across age groups?

- How does employee tenure differ between attrition groups?
- Which departments show different levels of job satisfaction?
- How is the workforce distributed across departments?

The dashboard therefore moves beyond simply displaying employee counts and provides a framework for investigating workforce behavior.

6.8 BUSINESS RECOMMENDATIONS

Based on the dashboard structure and analysis, the following actions can support HR decision-making:

Monitor High-Attrition Areas

Departments and job roles showing comparatively higher attrition should be investigated further to identify possible organizational causes.

Review Overtime Patterns

HR teams can examine overtime-related attrition patterns and assess whether workload or work-life balance may require attention.

Strengthen Employee Retention

Employee groups showing higher turnover can be considered for targeted retention initiatives.

Monitor Job Satisfaction

Department-level satisfaction should be monitored regularly to identify areas where employee experience may require improvement.

Analyze Employee Tenure

Understanding how attrition relates to organizational tenure can help HR teams design appropriate onboarding, development, and retention strategies.

Use Interactive HR Reporting

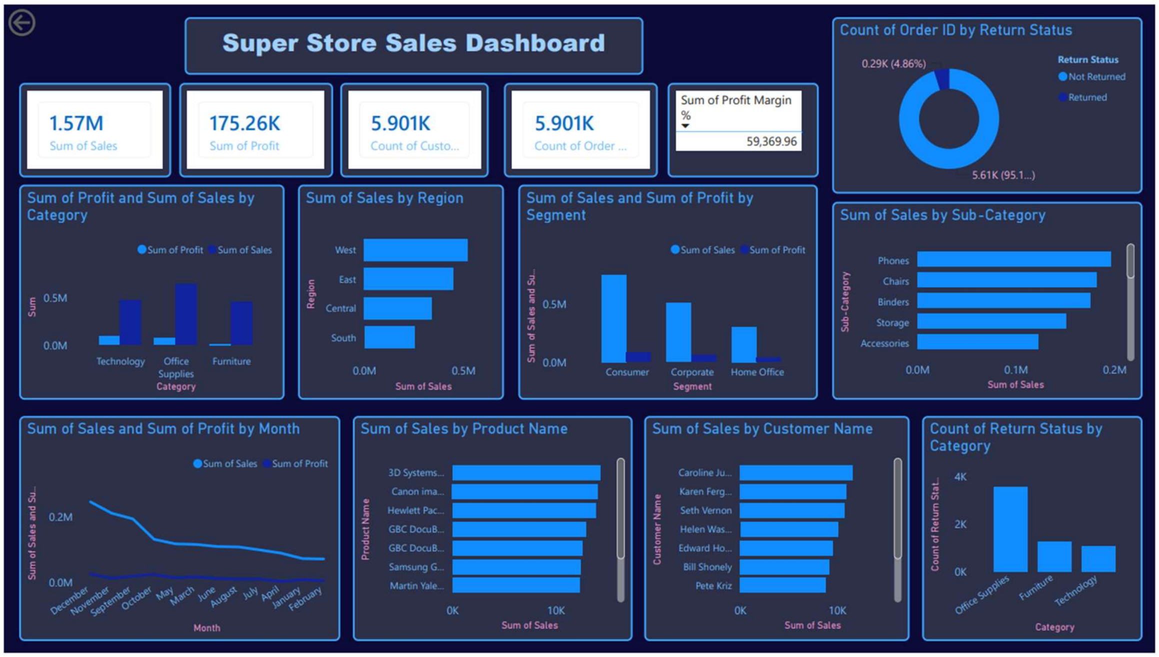
An interactive dashboard allows HR stakeholders to investigate workforce patterns without requiring separate reports for every employee segment.

7. COMPARISON OF THE THREE PROJECTS

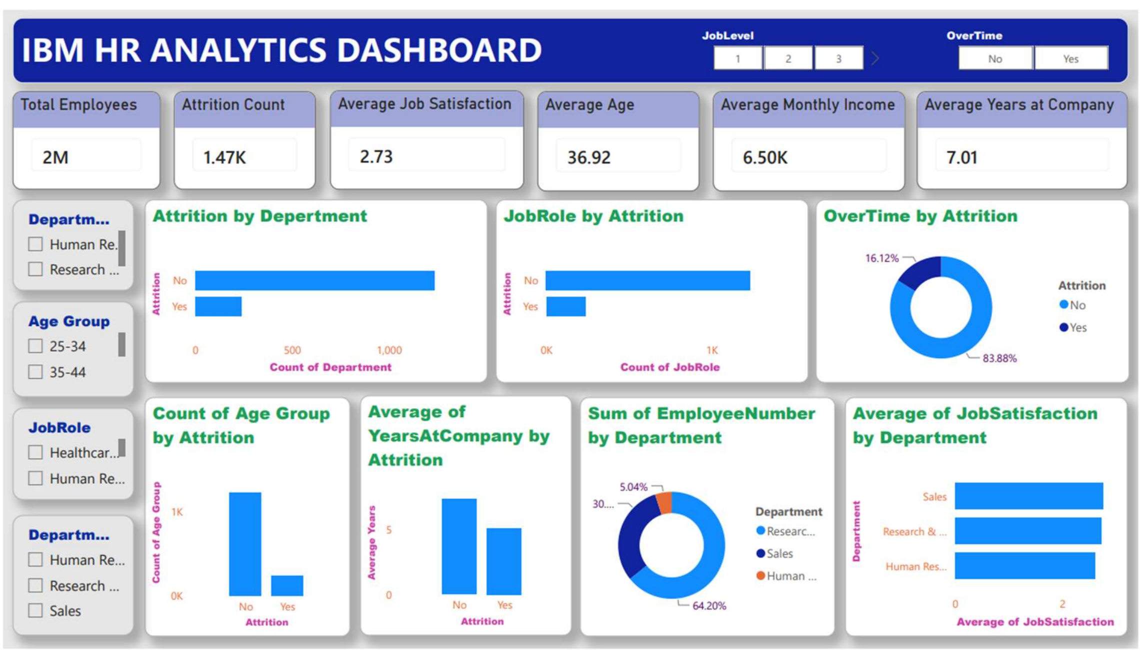
Area	Beginner	Intermediate	Advanced
Dataset	Vrinda Store	SuperStore	IBM HR Analytics
Primary Tool	Excel	Python + Power BI	Python + Power BI
Data Cleaning	Basic	Structured	Structured + Transformation
Visualization	Excel Charts	Power BI	Executive Power BI
Interactivity	Slicers	Slicers	Slicers + KPI-driven dashboard
Analysis	Sales	Sales & Business Performance	HR & Workforce
Business Focus	Basic	Intermediate	Executive
Reporting	Dashboard + Report	Dashboard + Report	Executive Dashboard + Report

Vrinda Sales Dashboard – Beginner:

Super Store Dashboard – Intermediate



IBM HR Analytics Dashboard – Advance



8. SKILLS AND LEARNING OUTCOMES

8.1 Data Cleaning

I gained practical experience in inspecting datasets, identifying potential data-quality issues, validating fields, and preparing datasets for analysis.

8.2 Exploratory Data Analysis

The projects demonstrated how different dimensions of a dataset can be explored to identify trends, differences, and patterns.

8.3 Data Visualization

I learned that effective visualization is not simply about creating more charts. The choice of visual should be connected to the question being answered.

8.4 KPI Development

The Advanced project particularly strengthened my understanding of executive-level KPI selection and dashboard hierarchy.

8.5 Dashboard Development

The projects provided experience in designing dashboards that combine:

- Summary metrics
- Charts
- Filters
- Comparative analysis
- Business context

8.6 Business Storytelling

One of the most important learning outcomes was understanding how to transform analytical results into a structured business story.

The official Advanced requirements emphasize that the dashboard should resemble a real business dashboard rather than simply being a collection of charts and should help stakeholders quickly understand what is happening and what decisions may be required.

8.7 Git and GitHub

The internship also provided practical experience in:

- Repository organization
- Folder structures
- Git commits
- Git branches
- Remote repositories
- Uploading project deliverables

The completed work is organized into separate Beginner, Intermediate, and Advance folders.

9. CHALLENGES AND PROBLEM-SOLVING

Dataset Preparation

Different datasets required different preparation approaches. Rather than applying the same cleaning process to every dataset, the preparation was adapted according to the structure and requirements of each dataset.

Dashboard Design

Creating an effective dashboard required decisions regarding:

- Which KPIs to display
- Which charts were necessary
- How to arrange visuals
- Which filters would be useful
- How to avoid unnecessary visual clutter

This was particularly important in the Advanced project, where the objective was to create an executive-style dashboard.

Analytical Progression

Moving from Excel-based analysis to Python and Power BI required learning new tools and changing the approach to data analysis.

The progression helped build confidence in handling datasets programmatically and presenting the results interactively.

GitHub Management

Managing project files through Git introduced practical challenges involving commits, remote changes, staged files, and repository synchronization.

Working through these issues provided valuable experience in version control and project organization.

10. OVERALL BUSINESS AND ANALYTICAL APPROACH

Across the three projects, a consistent analytical workflow was developed:

Raw Dataset

↓

Data Inspection

↓

Data Cleaning / Preparation

↓

Exploratory Analysis

↓

Identify Business Questions

↓

Select KPIs and Visuals

↓

Build Dashboard

↓

Interpret Results

↓

Business Insights

↓

Recommendations

This workflow demonstrates an important principle of data analytics: the objective is not simply to produce charts, but to use data to answer meaningful questions.

The three projects gradually developed this approach.

The Beginner project established the fundamentals of data summarization and visualization.

The Intermediate project introduced structured data preparation and deeper business analysis.

The Advanced project brought these skills together into an executive-style interactive dashboard focused on workforce decision-making.

11. CONCLUSION

The ShadowFox Data Analytics Internship provided a progressive practical learning experience covering three levels of data analytics.

The Beginner Level Vrinda Store project established foundational skills in Excel, Pivot Tables, charts, slicers, and basic sales analysis.

The Intermediate Level SuperStore project expanded these skills by introducing Python and Pandas for data preparation, exploratory analysis, and Power BI dashboard development.

The Advanced Level IBM HR Analytics project further developed the ability to create an executive-style dashboard using KPIs, interactive filters, workforce analysis, and business-focused visual storytelling.

Overall, the internship strengthened my ability to work through the complete data-analysis process—from raw data preparation to visualization, interpretation, and recommendations.

The most important learning was that successful data analytics requires more than technical tools. It requires understanding the business question, selecting relevant metrics, presenting information clearly, and converting analytical findings into information that can support decisions.

Through these three projects, I developed a stronger practical foundation in Excel, Python, Pandas, Power BI, data visualization, dashboard design, KPI development, business analysis, and GitHub-based project management.

The completed projects collectively demonstrate my progression from basic data analysis toward more structured and professional business intelligence reporting.